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4 A mass damper can be used to stabilize a building during earthquakes. A mass-spring system shown in Fig 4.1 can be used to model a mass damper.

A 600 g mass is placed on a smooth surface and is attached horizontally to two unstretched identical springs X and Y with a spring constant of 20 N m^{-1}.

When the mass is displaced by 5.0 cm from its equilibrium position and released from rest as shown in Fig. 4.2, it undergoes simple harmonic motion.

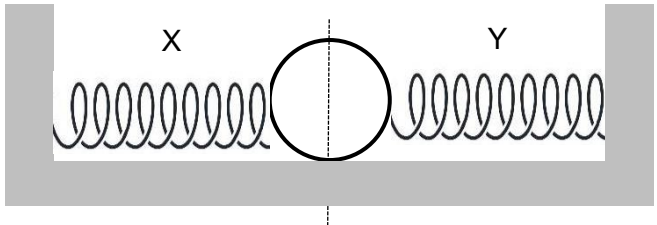

Fig. 4.1

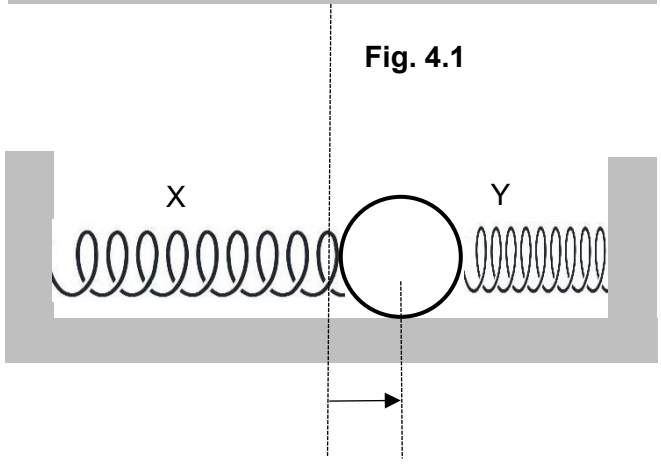

Fig. 4.2

(a) **(i)** Using energy consideration, calculate the maximum velocity of the mass.

maximum velocity = ……. m s^{-1} [3]

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(ii) Hence, determine the period of the oscillating spring-mass system.

period = ……. s [2]

(iii) Sketch on Fig 4.3 the variation with time *t* from the point of release of the mass

1. of the kinetic energy of the mass. Label this graph M.

2. of the elastic potential energy of spring Y. Label this graph S.

Show 1 complete oscillation. [2]

Fig. 4.3

(b) Suggest how the mass damper can help stabilize a building during an earthquake.

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…… [1]